



Seven Stages of Data Visualization Transcript

Hello everyone, welcome to BITS PLANET digital online programme. Today I will be taking you through the lesson 2 as part of our data visualisation and storytelling course. This lesson is part of module 1 which is introduction to visualisation.

In this lesson we will be taking care of two important aspects of building visualisations. One is the seven steps of data visualisation and how to choose different chart types for the purpose at hand. So this is an introductory session in continuation to the last lesson.

We will be seeing the basic elements of building a visualisation and the process of using different visuals for conveying our messages. So what are the seven stages of data visualisation? Whenever you embark on a data visualisation exercise, there is some pre ground work that you have to do before you build a visual and ensure that it is effectively delivering your message. So we are following the established principles of visualisation in which seven stages have been identified for developing a visual.

The first step of any data visualisation exercise is acquiring data. You go to the source or to the file in which the data for which you want to visualise is present and acquire the data. Most of the times the data is not readily available, therefore to create a visualisation you might have to visit multiple sources or collect the data that you need for your purpose.

In our case the data that we use for our example is coming from Election Commission of India website. We have provided the links to that and you can download the data from that website and use it for our exercises. Like in this case you also have other sources of data which you can download and experiment with.

Then you have parsing. Parsing is where you try to understand the structure of the data, what are its components, what are the different variables that are present, you just eyeball about checking about data integrity whether there are any gaps, whether it is text data, whether it is numerical data or combination of both. So you parse, skim through the data.

Then you filter the data based on preliminary understanding on what acts as a filter or what is a major attribute. There might be 10 or 12 attributes in your data or variables that you are interested in. You might want to reorient the entire data based on one particular filter.

A simple example would be gender. In any data set if you have gender you can simply divide it into two or three binary, two or three classes depending upon number of classes in the gender not just male and female and other classes if they are there and you can understand what is the behaviour of the variables based on this particular filtering. Filtering enables you to create partitions in the data that you think are important and filtering can be done on multiple variables and also depending upon the case.

Then once you filter you try to understand the patterns in the data that is where you are doing mining. You are going deeper right from acquiring data to seeing the components through parsing and filtering them into different categories of subspaces or subsets. Then you are trying to mine the data for uncovering patterns in the data.

So this is a very important step where you are in some sense performing an exploratory analysis to understand what is the data trying to talk to you and once you have formed a basic understanding you would then represent the data using simple visualisations thinking what is you saw. In the mining stage you might have found certain patterns you try to represent them using very simple visuals. Then finally when you go with this simple representations to your stakeholders audience or the people who want whom you want to see your visualisation you refine them to build more insights into it.

That is the phase where you refine the visualisations and finally you interact with your stakeholders. After mining the data you would have observed certain patterns that you want to showcase and once you want to showcase these patterns you would begin with representing them with very simple visualisations be it a bar chart or a distribution or a pie diagram. They might not completely represent what you want to showcase but this is a starting point.

Once you build a visual you refine them you try to build more context using just like principles or you want to bring in more additional information then try to present even more a better visual. This is where the refining of the visual happens. Once you refine the visual finally you will go back and interact with your stakeholders the audience to whom you want to convey this story and based on their feedback or their questions you might come back directly again to move refine the visualisation and present it in a better fashion or at times you might have to take a deeper look at the data that is when from interaction you while directly go back to the mining stage or you can go back from refinement to filtering adding more categories to partition the data or finally at times you might not have the data that the your stakeholders are asking so you go to the source and add a new data source or a new data element.

So this process is not just iterative but it is also sequential it moves parallel sometimes but these are seven broad steps which every data visualisation exercise will go through. So these seven steps are essential for every data visualisation exercise sometimes they run sequentially sometimes they run parallelly and most of the times they run iteratively. So data developing a data visualisation is an iterative process starting from the acquisition of data parsing it filtering it out mining it and representing refining and finally interacting with the stakeholders and doing the loop again at different levels.

So keep this in mind while you are developing your data visuals. So let us take an example and closely look at this visualisation exercise. These seven stages of visualisation exercise as I said for this example we are taking data from election commission of India the motor turnout on data once we acquire one caveat here or a

caution whenever you are sourcing data you have to be very careful of the format whether it is in Excel CSV or whether it is more is most importantly.

Let me start off with an example this represents the acquiring stage of our data visualisation. A word of caution whenever you are acquiring data you have to ensure the compatibility of the data with your system. For example you have a software which does not accept Excel files or CSV files there is no point in downloading data in that format or if you have data in a particular format you might have to convert it into an amenable format for your software.

These checks have to be made at the acquiring stage so that you are able to do your data visualisation process in a consistent and smooth manner okay without much of frictions. So we are operating on Excel so the data is also available in the Excel format we have just downloaded it and now we come to the parsing stage. This is the how the data would look with the different columns so in the parsing stage you would take a look at the data and its components.

What are the major components in this data? You have the state name, you have the number of electors including the service electors so total polling stations in the state and the number of voters who have casted vote through normal electronic voting machines at EVMs. Then you have postal ballots, total votes and the voter turnout ratio at the last column. The main variable of interest is voter turnout ratio and we have seen that and what we can do is simply see the other parameters for us they are not immediately relevant so we are just looking at the voter turnout ratio and seeing that as the main variable of interest.

Then we will go to filter as we have identified voter turnout ratio as the main variable of interest we are going to sort or filter our data based on voter turnout ratio. In this step I have filtered the data in descending order starting from highest to lowest resultantly Lakshadweep the state with highest voter percentage is at the top Bihar with the lowest voting share is at the bottom and you might notice All India is coming in between with All India voting share as 66.1% though All India can come at the bottom as total because you have just filtered the data it has come in the middle. This highlights the filtering stage of the visualisation.

Again for us the main point is on what parameter you are trying to filter you can as well filter on polling stations but that does not give much of a sense to the data that we are looking at we are looking at voter turnout participation so our major variable of focus for filtering is voter turnout ratio. Just for a check I have used gestured principles of similarity and coloured the states which are above national average in one particular colour that is blue and the ones which are below national average that is light brown. So you can use gestured principles across table I mean across different visualisations to convey information at different stages of developing visuals also.

So next we come to the stage of mining the data. In this we have already filtered the data in the descending order of voter turnout ratio and we identified I mean I am not

presenting the entire excel sheet or the data frame here you can do that at your end but once we count the number of states that are above national average is 23 below is 13 so there is an all India benchmark these aspects are available at the mining stage. This is where you try to understand how a particular data point that is voter turnout ratio is associated with other parameters and what is the relation that we found we found number of states which are above all India average and the states which are below all India average.

This is the mining stage now you can also do number of voters right number of voters as a percentage of polling stations you can also create a metric you can do some other transformations here we are not getting into that but this is where you would explore the data for different patterns that is what mining is a major part of data visualisation exercise is data mining where you try to uncover the hidden patterns right in this case we are restricting ourselves to voter turnout ratio but as we go through the course we would be doing lot of mining to build effective visuals. Then finally we do a simple representation of the data we start with alphabetical order and present the voter turnout ratio and then just present the horizontal bar graph by the looks of it one can understand this is not a very effective visual it is not conveying what we have observed in the mining stage. The mining stage has clearly brought out there are 23 states which have more than average voter turnout ratio and 13 states have less than average voter turnout ratio.

Resultantly this graphic is not conveying that information it is blunt it is just showing the state and their voting turnout ratio without conveying the information that we have observed in the mining stage. This is where we go refine the data and try to present the story that we want to convey using the visual. We have transformed the visual from simple horizontal bar graph to ordered horizontal bar graph by taking into account the descending order of the voting ratio.

So how did we do this is where you refine your graphics to convey the message that you have found in the mining stage. The first graph is simple though it uses some of the just like principles like for form and ground and tries to highlight the background it is not able to convey the message that we are trying to convey. So what did we do we form a natural ordering of the data that is voter turnout ratio which is descending to ascending highest to lowest we start with Lakshadweep and we go all the way down to Bihar.

Then we also have a national average line which is the principle of focus. So you try to show that this is the line you can simply see in the entire graphic this line has a distinct focus and the states above the line have higher voter average below the line have lower voter average and if somebody can count it is 13 and 23 and the highest is Lakshadweep lowest is Bihar it is conveying all the messages that we want to convey in a way it is a very good graphic that we want to take to our audience. Once we take this to our audience there might be other questions right.

So there might be questions on tell me which is the particular state which is in the highest and which is particularly in the lowest try to answer these questions by refining it. So you try to in this case highlight Lakshadweep in a different colour, highlight Bihar in a different colour to show that they are the states with highest and lowest voter share. Then there might be a question what is the region wise performance.

So you can go add so now when we talk from a data visualisation perspective you do not have region in the original data set there might be multiple reasons for that the elections are conducted in phases maybe they want to look at different regions for a reason. So you do not have that particular data you would go for up you directly enter from interaction to acquiring stage. So you go add a variable called region in the data and then present the entire thing in the graphic.

So once the questions get asked at the interaction stage you might have to go back to the acquisition stage or do a mining do a representation and refinement to present your visual. These are the seven steps of data visualisation almost all visualisations that you build through go through all this process. Sometimes when you are using automated software these steps might not be visible to you or even not recognise them, but effectively these are there at the back end some software is taking care of you, but be aware of these seven steps and know where to intervene to improve the effectiveness of your visual.

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